Math Inquirers

Extension Workbook

with answers

Level 4

Self-study | Ages 9-11



My name _____

About this book

Math Inquirers Extension Workbook, Level 4
Self-study practice with answers.

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Good to know

- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

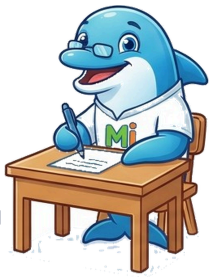
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

Contents

	About this book Copyright and licence	page 2
	Welcome	page 3
	My pace plan	page 5
1	**Factor hunters** Not in this proof edition	–
2	**Big number puzzles** Not in this proof edition	–
3	**Operation detectives** Not in this proof edition	–
4	**Fraction frontier** Not in this proof edition	–
5	**Decimal detectives** Not in this proof edition	–
6	**Angle turns** Not in this proof edition	–
7	**Area architects II** Not in this proof edition	–
8	**Symmetry and Nets** Symmetry and nets · follows Holiday Program Lesson 8 · answers on page 16	page 6
9	**Data detectives** Not in this proof edition	–
10	**Puzzle day** Not in this proof edition	–
	My progress	page 13
	My certificate	page 14
	Answers Check your work, with short reasons	page 15

My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Factor hunters	Not in this proof edition				☐
2	Big number puzzles	Not in this proof edition				☐
3	Operation detectives	Not in this proof edition				☐
4	Fraction frontier	Not in this proof edition				☐
5	Decimal detectives	Not in this proof edition				☐
6	Angle turns	Not in this proof edition				☐
7	Area architects II	Not in this proof edition				☐
8	Symmetry and Nets	6-7	8-9	10 and 12	11	☐
9	Data detectives	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

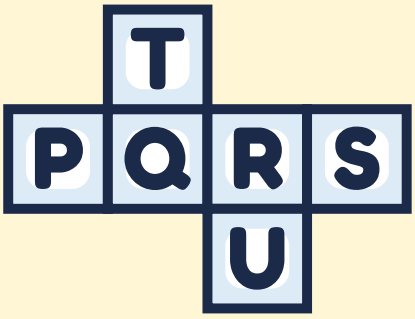
Learn it: Symmetry and nets

My goal. I can tell which nets fold into a cube and which faces are opposite. I can complete a figure so that a line is a line of symmetry.

A net is a flat shape that folds into a solid. A cube net has 6 squares. In a line of 3 squares, the 2 end squares are opposite faces. On a dice, opposite faces add up to 7.

Worked example 1

Does this net fold into a cube? Which face is opposite P?

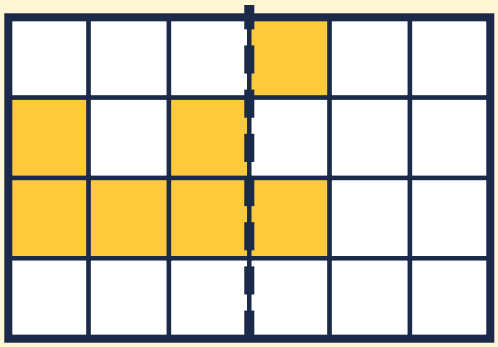


1. Choose a base. The row of 4 squares wraps round as walls.
2. One extra square is above the row and one is below, so they make the lid and the base.
3. P and R are 1 square apart in the row, so they are opposite.

Answer: It folds into a cube. R is opposite P.

Worked example 2

Complete the figure so that the dashed line is a line of symmetry.



1. Flip each shaded square across the line.
2. A square 1 away on one side has a partner 1 away on the other side.

Answer: Each shaded square gets a partner the same distance from the line.

How to do it

1. Net: choose a base. Fold the squares round it in your head.
2. A net needs 1 base, 1 lid and 4 walls.
3. Symmetry: count squares from the line to the partner.

Good to know

- Dice: opposite faces add up to 7.
- A rectangle has 2 lines of symmetry. A square has 4.

Your turn

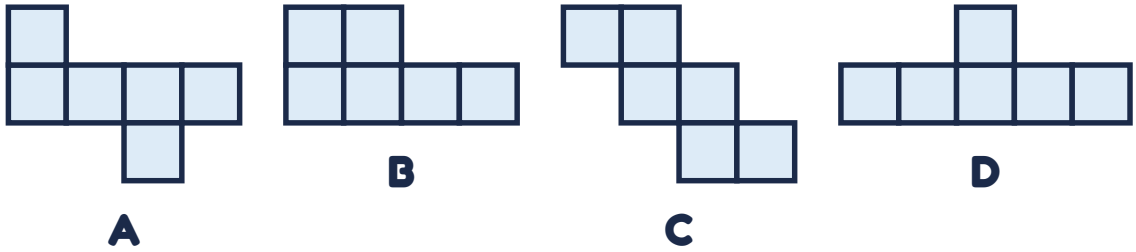
Answer at the back of the book.

Try **Opposite faces of a dice add up to 7. What number is opposite 4?** ☐

Start here

Do these first. Take your time.

1 **Which of these nets fold into a cube? Circle the letters.**



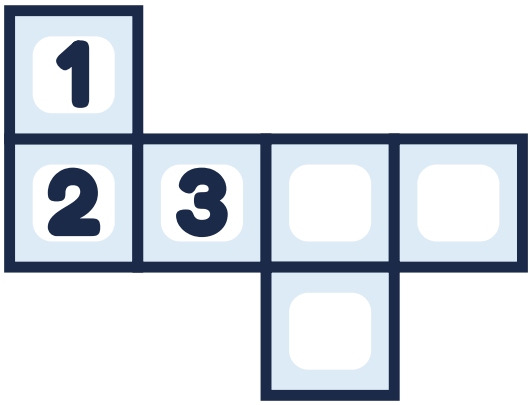
2 **This net folds into a cube. Which letter is opposite P? Q? S?**



(a) Opposite P: ☐ (b) Opposite Q: ☐

(c) Opposite S: ☐

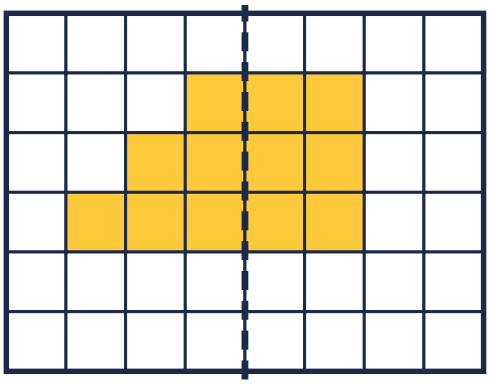
3 **Opposite faces of a dice add up to 7. Write the missing numbers on the net. Go row by row.**



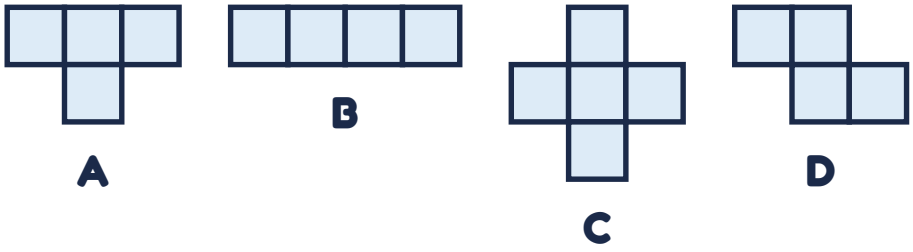
Start here

Keep going. Check your work.

4 Shade squares to complete the figure so that the dashed line is a line of symmetry. How many squares are shaded when it is complete? ☐

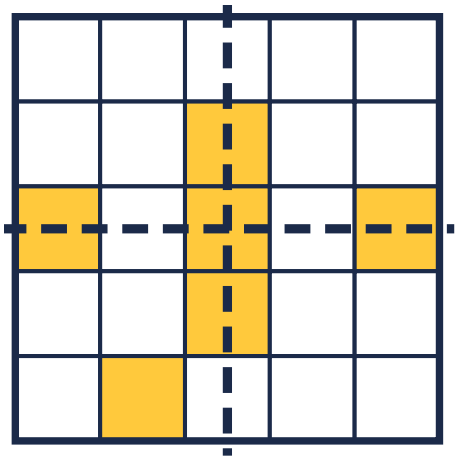


5 How many lines of symmetry does each shape have?



(a) A: ☐ (b) B: ☐ (c) C: ☐ (d) D: ☐

6 Shade ONE more square so that one of the dashed lines is a line of symmetry of the shaded squares.



(a) How many different squares can you shade? ☐

Start here

Nearly there. You can do it.

7 **A dice sits on a table. The top face shows 6 and the face in front shows 4.**

(a) The bottom face shows ☐

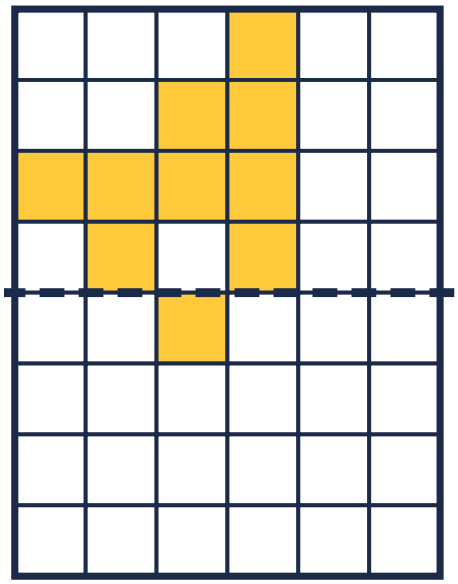
(b) The face at the back shows ☐

(c) The 4 side faces add up to ☐

8 **A box is 4 cm long, 3 cm wide and 2 cm high. Its net is cut from card. What is the area of the card?** ☐ **cm²**

Working: _____

9 **Shade squares to complete the figure so that the dashed line is a line of symmetry. How many squares are shaded when it is complete?** ☐



10 **A cube has edges of 5 cm.**

(a) All 12 edges are 5 cm long. The edges add up to ☐ cm

(b) The area of its net is ☐ cm²

Think harder

Hints are upside down. Try first!

1 **A cube has faces numbered 1 to 6. Face 1 touches faces 2, 3, 5 and 6. Face 2 touches faces 1, 3, 4 and 5.**

(a) Opposite 1 is ☐

(b) Opposite 2 is ☐

(c) Opposite 3 is ☐

Working: _____

Hint. A face touches 4 faces and is opposite the 1 face it does not touch.

2 **Mei looks at a dice from a corner and sees 3 faces. The numbers on them add up to 10.**

(a) The numbers she can see, smallest first: ☐ ☐ ☐

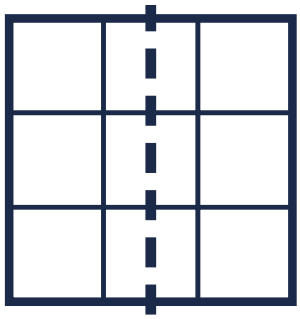
(b) The numbers she cannot see, smallest first: ☐ ☐ ☐

Working: _____

Hint. At a corner, the 3 faces are from 3 different opposite pairs.

3 **On a 3 by 3 grid of squares, shade exactly 3 squares so that the up-down line through the middle is a line of symmetry. How many different patterns can you make?**

☐



List: _____

Hint. Squares in the middle column are their own partners. Others come in pairs.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 Beyond Level 4: Painted cubes

A 3 by 3 by 3 cube is painted on all 6 outside faces. Then it is cut into 27 small cubes.

(a) Small cubes with 3 painted faces: ☐

(b) Small cubes with 2 painted faces: ☐

(c) Small cubes with 1 painted face: ☐

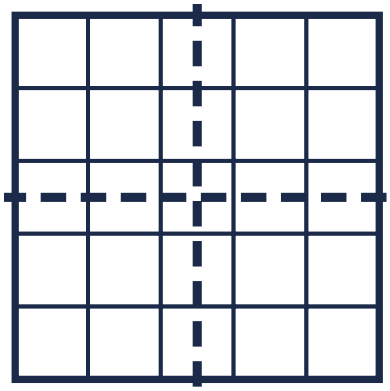
(d) Small cubes with no paint: ☐

Hint. Corners, edges and faces of the big cube. Count each kind.

2

On a 5 by 5 grid of squares, shade exactly 5 squares so that the pattern has 4 lines of symmetry. How many different patterns are there?

☐



Hint. The middle square is always shaded. The other 4 squares go together.

3

A dice sits on a table with 1 on top, 2 facing north and 3 facing east. It is rolled over its north edge, then over its east edge, then over its north edge again. When it rolls north, the face that was on its north side lands on the table.

(a) The top face is now ☐

(b) The bottom face is now ☐

Top, north, east: _____

_____

_____

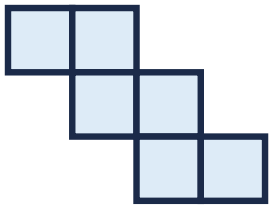
Hint. Write the top, north and east faces after each roll.

✔ Check yourself

No hints. Try each one on your own.

1

Does this net fold into a cube? **Yes** **No**



2 **How many lines of symmetry has a rectangle that is not a square?** ☐

Working: _____

3 **A dice has 2 on top and 3 in front. The bottom shows** ☐ **and the back shows** ☐ **.**

Working: _____

4 **A cube has edges of 4 cm. Its net has an area of** ☐ **cm².**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Factor hunters	☐	☐	☐	☐	/ 4	
2	Big number puzzles	☐	☐	☐	☐	/ 4	
3	Operation detectives	☐	☐	☐	☐	/ 4	
4	Fraction frontier	☐	☐	☐	☐	/ 4	
5	Decimal detectives	☐	☐	☐	☐	/ 4	
6	Angle turns	☐	☐	☐	☐	/ 4	
7	Area architects II	☐	☐	☐	☐	/ 4	
8	Symmetry and Nets	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 4.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 8 · Symmetry and Nets

Your turn
page 6

Answer: 3
Why: 4 + 3 = 7, so 3 is opposite 4.

START HERE

1
page 7

Answer: A and C
Why: Fold each net in your head. A: the row of 4 makes the walls, and one square above and one below make the lid and the base. It folds. B: the 2 by 2 block of squares has 4 squares meeting at a corner, but only 3 faces of a cube meet at a corner. D: 5 squares in a row wrap round and the 5th covers the 1st. C: the staircase folds into a cube.

2
page 7

Answer: (a) R **(b)** T **(c)** U
Why: In the row of 3 squares P, Q, R, the 2 ends P and R are opposite. Q and T are opposite (1 square between them going round the step). The last pair is S and U.

3
page 7

Answer: 5, 4, 6 (row by row)
Why: Find the opposite squares. 2 and the third square of the row are opposite, so that square is 7 − 2 = 5. 3 and the fourth square are opposite, so it is 7 − 3 = 4. 1 and the bottom square are opposite, so it is 7 − 1 = 6.

4
page 8

Answer: 24
Why: There are 12 shaded squares on the left of the line. Each needs a partner on the right, so 12 more squares are shaded. 12 + 12 = 24.

5
page 8

Answer: (a) 1 **(b)** 2 **(c)** 4 **(d)** 0
Why: A has 1 line (down the middle). B has 2 lines (along its length and across its middle). C, the plus sign, has 4 lines (up-down, across and the 2 diagonals). D has none: it looks the same when turned half way round, but no fold makes the halves match.

6
page 8

Answer: (a) 2
Why: Look for a shaded square with no partner. The square in the bottom-left area, 1 up from the bottom in column 2, has no partner across the up-down line: its partner is at the right in the same row. It also has no partner across the across line: its partner is at the top. Each choice fixes one dashed line, and no other square works. There are 2 ways.

7
page 9

Answer: (a) 1 **(b)** 3 **(c)** 14
Why: Opposite faces add up to 7. The bottom is 7 − 6 = 1. The back is 7 − 4 = 3. The 4 side faces are 2 opposite pairs, and 7 + 7 = 14.

8
page 9

Answer: 52
Why: The 6 faces are 2 of 4 by 3 (12 each), 2 of 4 by 2 (8 each) and 2 of 3 by 2 (6 each). 24 + 16 + 12 = 52 cm².

9
page 9

Answer: 20
Why: There are 10 shaded squares above the line. Each needs a partner below the line, so 10 more squares are shaded. 10 + 10 = 20.

10
page 9

Answer: (a) 60 (b) 150
Why: (a) 12 × 5 = 60 cm. (b) Each face is 5 × 5 = 25 cm², and the net has 6 faces: 6 × 25 = 150 cm².

THINK HARDER

1
page 10

Answer: (a) 4 (b) 6 (c) 5
Why: Face 1 does not touch 4, so 4 is opposite 1. Face 2 does not touch 6, so 6 is opposite 2. The faces left over are 3 and 5, so they are opposite.

2
page 10

Answer: (a) 1, 4, 5 (b) 2, 3, 6
Why: The pairs are 1 and 6, 2 and 5, 3 and 4. She sees one number from each pair, so no two of the 3 numbers add up to 7. The only choice with a total of 10 is 1, 5 and 4. The hidden faces are opposite: 6, 2 and 3.

3
page 10

Answer: 10
Why: The 3 shaded squares are 3 from the middle column, or 1 from the middle column and 1 pair. 3 in the middle column: 1 way. 1 in the middle column (3 choices) and 1 pair (3 rows to choose): 3 × 3 = 9 ways. 1 + 9 = 10.

GO FURTHER

1
page 11

Answer: (a) 8 (b) 12 (c) 6 (d) 1
Why: 3 painted faces: the 8 corner cubes. 2 painted faces: 1 cube in the middle of each of the 12 edges, so 12. 1 painted face: 1 cube in the middle of each of the 6 faces, so 6. No paint: the cube in the very middle, so 1. Check: 8 + 12 + 6 + 1 = 27.

2
page 11

Answer: 4
Why: With 4 lines of symmetry the squares come in groups. The middle square is 1 group. The other groups of 4 are the 4 corners, the 4 squares next to the corners along the diagonals, the 4 middle squares of the sides and the 4 squares touching the centre. One of these 4 groups plus the middle square makes 5 squares: 4 patterns.

3
page 11

Answer: (a) 6 (b) 1
Why: Start: top 1, bottom 6, north 2, south 5, east 3, west 4. Roll north: the north face 2 lands on the table, so the top is 5, the bottom is 2, the north face is 1 and the south face is 6. Roll east: the east face 3 lands on the table, so the top is 4, the bottom is 3, the east face is 5 and the west face is 2. Roll north: the north face 1 lands on the table, so the bottom is 1 and the top is the old south face, 6.

CHECK YOURSELF

1
page 12

Answer: Yes
Why: The staircase of 2, 2, 2 squares folds into a cube. Each step wraps round to make the base, walls and lid with no overlaps.

2
page 12

Answer: 2
Why: It can be folded along the middle in 2 ways, and the halves match. The diagonals do not work.

3
page 12

Answer: bottom 5, back 4
Why: Opposite faces add up to 7. The bottom is 7 − 2 = 5 and the back is 7 − 3 = 4.

4
page 12

Answer: 96
Why: Each face is 4 × 4 = 16 cm². The net has 6 faces: 6 × 16 = 96 cm².